Trend research — 2026-04-19 · incr 16

trending_swings is the only independent voter; one near-duplicate pair found

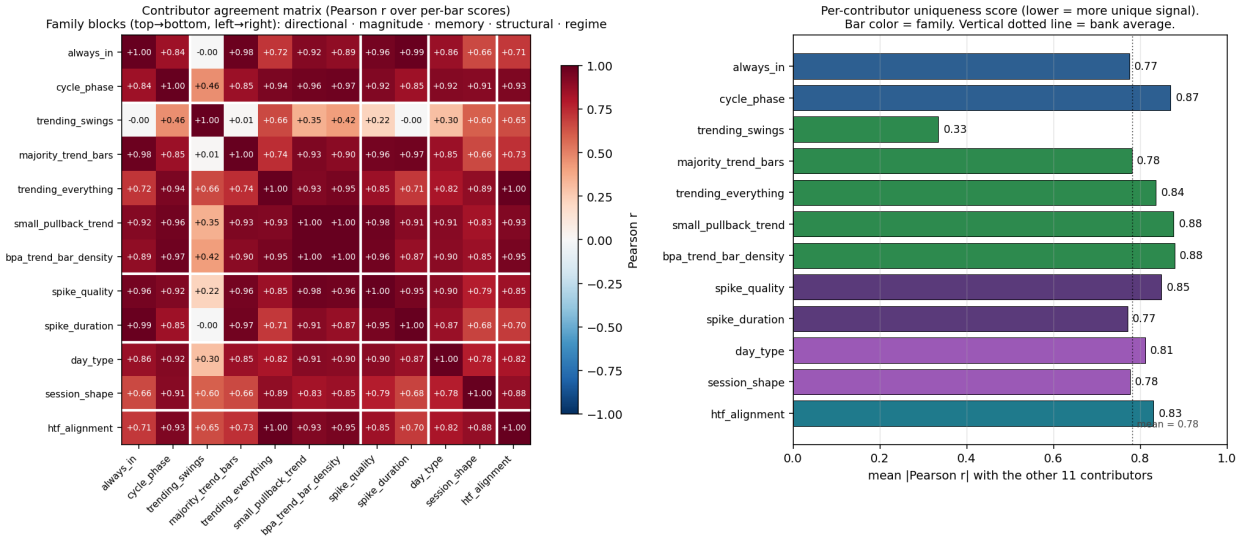
TL;DR

Computed pairwise Pearson correlation across all 12 trend contributors over a 130-row per-bar bank of the 5 canonical fixtures. Result: **11 of 12 contributors are silently double-counting each other** (mean |r| 0.77–0.88). Only **trending_swings** brings independent signal (|r|_avg = **0.34**). One near-duplicate pair found inside the directional family.

Headline findings

- **trending_swings** uniqueness **|r|_avg = 0.34** — less than half the next-best contributor and the only one below the bank average (0.79). Removing it would erase the only voter that uniquely fires on pullback sessions.
- **small_pullback_trend** ↔ **bpa_trend_bar_density** correlate at **r = +0.997** — near-duplicate inside the directional family. Strongest candidate to down-weight as a unit or drop one once Pattern Lab WRs decide.
- 11 of 12 contributors sit at **|r|_avg between 0.77 and 0.88**. Equal weighting is silently double-counting most of the stack.
- **Caveat:** several cross-family near-perfect r's (e.g. trending_everything ↔ htf_alignment = +0.995) are sample-bank artifacts of polarized synthetic fixtures + confluent HTF pairing. **Real-data validation is the next step before any weight change.**

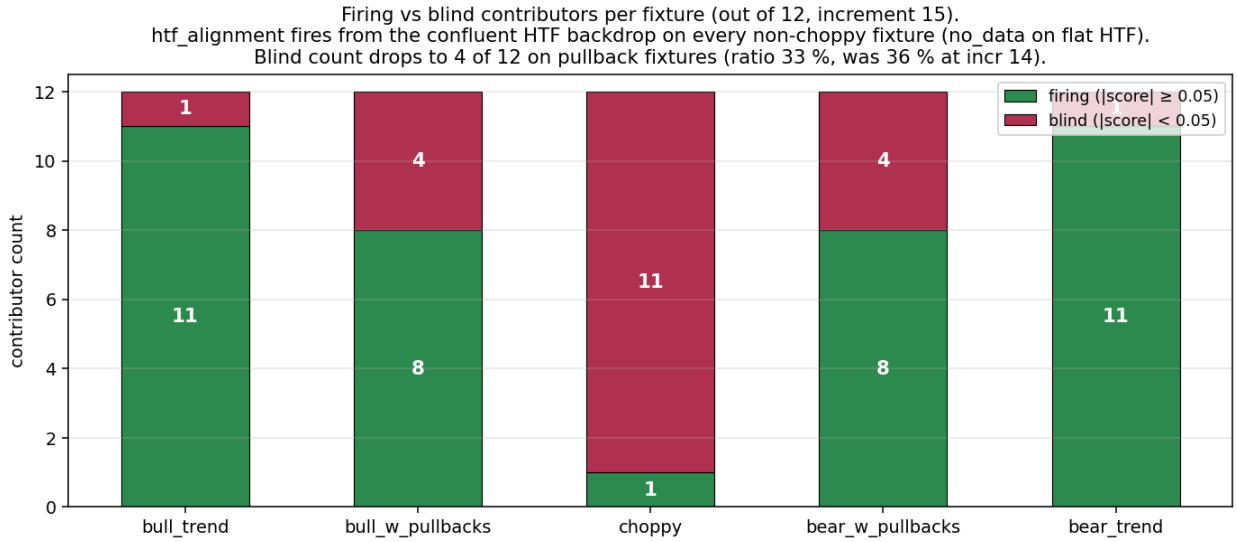
Contributor agreement matrix



Left: 12×12 Pearson r heatmap (red = move together, blue = opposite). Right: per-contributor uniqueness (mean |r| with the other 11). Lower bar = more independent signal. Dotted line is the bank average (0.79).

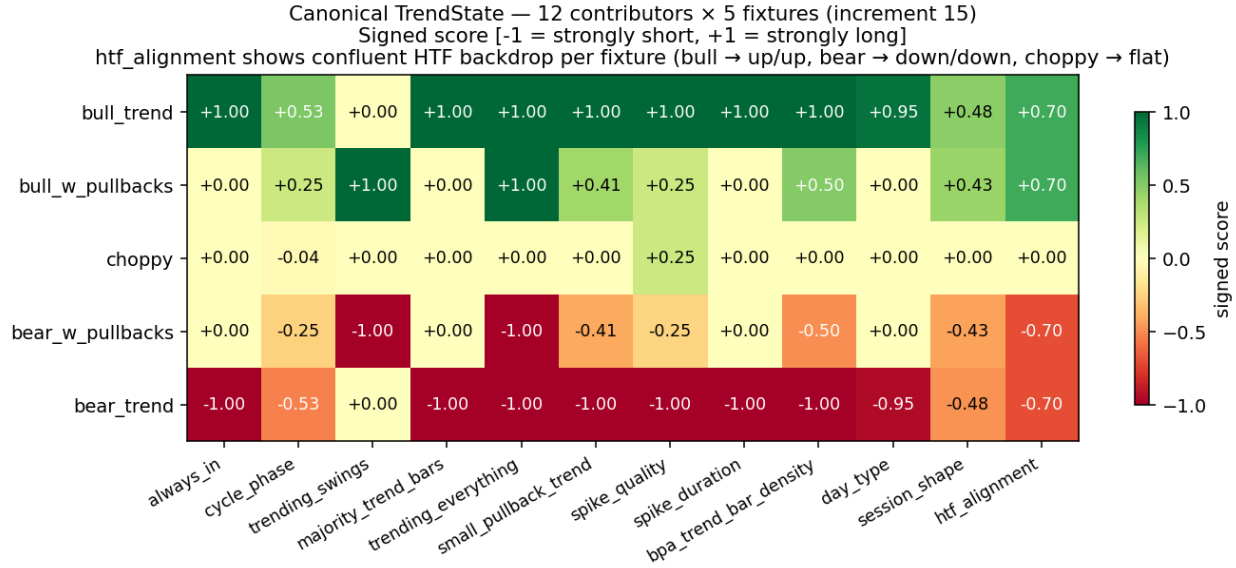
Why trending_swings matters — the blind-spot story

trending_swings is the only contributor that fires on pullback sessions and stays silent on monotonic trends — exactly opposite to most of the stack. The strict-threshold contributors (always_in, majority_trend_bars, spike_duration, day_type) go blind on the 4 of 12 pullback fixtures; trending_swings covers them.



Per-fixture firing vs blind contributors. trending_swings fills the gap left by the strict-threshold voters on pullback sessions.

The full 12 × 5 control panel



Each row is a canonical market regime; each column is one classifier's signed score. Dark green = strong long, dark red = strong short. Reads like a control panel for the whole study.

What's next — needs Will's nod

- **Emit trend_state into the dashboard payload.** Additive but visible.
- **Wire real daily/weekly close history into the pipeline** so htf_alignment isn't dormant in production.
- **Pattern Lab WR-driven weighting decision** — start by testing whether down-weighting the small_pullback_trend ↔ bpa_trend_bar_density near-duplicate pair improves WR.
- **Real-data redundancy validation** — re-run the incr-16 study on a backtest sample of real intraday sessions to confirm the synthetic-bank findings.

Findings site: https://github.com/zerosumsystems-ui/iPhone-/tree/main/trend-classification